

Homework #5. Due on Thursday, Oct 8th, 11:59pm on Canvas

Reading for this homework: Sections 2.5 and 2.6.

Plan for next week (Oct 5): 3.1 (elementary matrix operations and elementary matrices), 3.2 (the rank of a matrix) and (if time allows) parts of 3.4 (reduced row echelon form).

Problems:

Problem 1: Let $V = P_2(\mathbb{R})$ and define the linear map $T : V \rightarrow V$ by $T(f(x)) = (x + 1)f'(x)$.

- (a) Compute $[T]_\beta$ for the ordered basis $\beta = \{1, x, x^2\}$.
- (b) Let $\beta' = \{1, (x + 1), (x + 1)^2\}$. Compute the matrix $Q = [id_V]_{\beta'}^\beta$, which changes β' -coordinates into β -coordinates and then use the change of basis formula to compute $[T]_{\beta'}$.
- (c) Now compute $[T]_{\beta'}$ directly from the definition of T . **Hint:** If an element $f \in P_2(\mathbb{R})$ is given in the standard form ($f = ax^2 + bx + c$), to compute its β' -coordinates, let $y = x + 1$ (so that $x = y - 1$) and then expand f as a polynomial in y .

Problem 2: Recall the definition of a projection map introduced in Problem 5 on the first midterm. Let V be a vector space, and suppose $V = U \oplus W$ for some subspaces U and W . By Problem 1 on the practice midterm, every $v \in V$ can be uniquely written as $v = u + w$ with $u \in U$ and $w \in W$. Thus, we can define a map $P_{U,W} : V \rightarrow V$ by

$$P_{U,W}(u + w) = u \text{ for all } u \in U \text{ and } w \in W.$$

The map T (which is linear by Midterm 1.5(a)) is called the **projection onto U along W** and will be denoted by $p_{U,W}$. By a **projection** we will mean a map $p_{U,W}$ for some U and W as above.

Remark: $P_{U,W}$ is actually the unique linear map $T : V \rightarrow V$ such that $T(u) = u$ for all $u \in U$ and $T(w) = 0$ for all $w \in W$ (such T is unique since V is spanned by $U \cup W$).

Now assume that V is a finite-dimensional vector space and $T : V \rightarrow V$ a linear map. Prove that the following are equivalent:

- (i) T is a projection, that is, $T = P_{U,W}$ for some U and W with $U \oplus W = V$.
- (ii) $T^2 = T$ (where $T^2 = T \cdot T$ is the composition of T with itself)
- (iii) There is an ordered basis β of V s.t. $[T]_\beta = e_{11} + \dots + e_{kk}$ for some $0 \leq k \leq \dim V$, that is, $[T]_\beta$ is the diagonal matrix whose first k

diagonal entries are equal to 1 and the remaining diagonal entries are equal to 0.

The equivalence of (i) and (iii) is Problem 5(b) on the first midterm. If you solved this problem correctly, you do not have to redo this part.

Hint: Perhaps the shortest way to solve Problem 2 is via the cycle “(i) $\Rightarrow$ (iii) $\Rightarrow$ (ii) $\Rightarrow$ (i)”. The first two implications are relatively easy. To prove “(ii) $\Rightarrow$ (i)” first show that if $T^2 = T$, then $V = \text{Im}(T) \oplus \text{Ker}(T)$.

Problem 3: Let V be a finite-dimensional vector space over a field F and W a proper subspace of V .

- (a) Suppose we are given a linear map $g : W \rightarrow U$ for some vector space U over F and vectors $v \in V \setminus W$ and $u \in U$. Prove that there exists a linear map $f : V \rightarrow U$ such that $f = g$ on W (that is, $f(w) = g(w)$ for all $w \in W$) and $f(v) = u$. **Hint:** Apply Lemma 5.1 from class to a suitable basis of V .
- (b) Use (a) to show that there exists a nonzero linear map $f : V \rightarrow F$ (that is, $f \in V^*$) such that $f|_W = 0$ (where as before $f|_W$ is the restriction of f to W)
- (c) If $f \in V^*$, the map $f|_W : W \rightarrow F$ is also linear (so lies in W^*). Thus, we have a natural map $\varphi : V^* \rightarrow W^*$ given by $\varphi(f) = f|_W$. Prove that φ is linear, and then deduce from (a) and (b) that φ is surjective and not injective.

Problem 4: This problem defines the notion of a direct sum of more than two spaces. As with the case of 2 subspaces, we will define both external and internal direct sums, but this time we start with the definition of the external sum, which is a straightforward generalization of the case of 2 spaces.

Let $V_1, \dots, V_n$ be vector spaces over the same field F . The external direct sum $\oplus_{i=1}^n V_i$ is defined as the set of all n -tuples $\{(v_1, \dots, v_n) : v_i \in V_i \text{ for } 1 \leq i \leq n\}$ with componentwise addition and scalar multiplication.

Now assume that $V_1, \dots, V_n$ are subspaces of the same space V . Their internal sum $\sum_{i=1}^n V_i$ (not necessarily direct) is defined as

$$\sum_{i=1}^n V_i = \{v \in V : v = \sum_{i=1}^n v_i \text{ for some } v_i \in V_i\}.$$

As in HW #4.1, there is a natural linear map $T : \oplus_{i=1}^n V_i \rightarrow \sum_{i=1}^n V_i$ (where the sum on the left is the external direct sum) given by

$$T((v_1, \dots, v_n)) = \sum_{i=1}^n v_i.$$

Prove that the following are equivalent:

- (a) T is an isomorphism
- (b) $\text{Ker}(T) = \{0\}$
- (c) For all $v \in \sum_{i=1}^n V_i$ there exist UNIQUE vectors $v_1 \in V_1, \dots, v_n \in V_n$
s.t. $v = \sum_{i=1}^n v_i$
- (d) $V_i \cap (\sum_{j \neq i} V_j) = 0$ for all $1 \leq i \leq n$
- (e) $V_i \cap (\sum_{j=1}^{i-1} V_j) = 0$ for all $2 \leq i \leq n$

We say that the sum $\sum_{i=1}^n V_i$ is direct and write $\oplus_{i=1}^n V_i$ if one (hence also all others) of the above five conditions hold.

Problem 5: Give an example of a vector space V and three subspaces W_1, W_2, W_3 of V such that $W_i \cap W_j = \{0\}$ for all $i \neq j$, but the sum $W_1 + W_2 + W_3$ is not direct.

Problem 6: In each of the following examples you are given a finite-dimensional vector space V and a basis $\beta = \{v_1, \dots, v_n\}$ of V . Find the dual basis $\beta^* = \{v_1^*, \dots, v_n^*\}$ and give an explicit formula for each element of β^* . For instance, in part (a) the answer should be given in the form $v_1^*((a, b)) = \text{some explicit function of } a, b$ and similarly for v_2^* .

- (a) $V = \mathbb{R}^2$, $\beta = \{(3, 4), (7, 8)\}$
- (b) $V = P_1(\mathbb{R})$, $\beta = \{x + 1, x + 2\}$.

Problem 7: Let $V = P_2(\mathbb{R})$ and consider the elements f_1, f_2, f_3 of V^* given by $f_1(a+bx+cx^2) = a+b$, $f_2(a+bx+cx^2) = b+c$ and $f_3(a+bx+cx^2) = a+c$. Prove that $\{f_1, f_2, f_3\}$ is a basis of V^* and find a basis β of V for which $\{f_1, f_2, f_3\}$ is the dual basis.